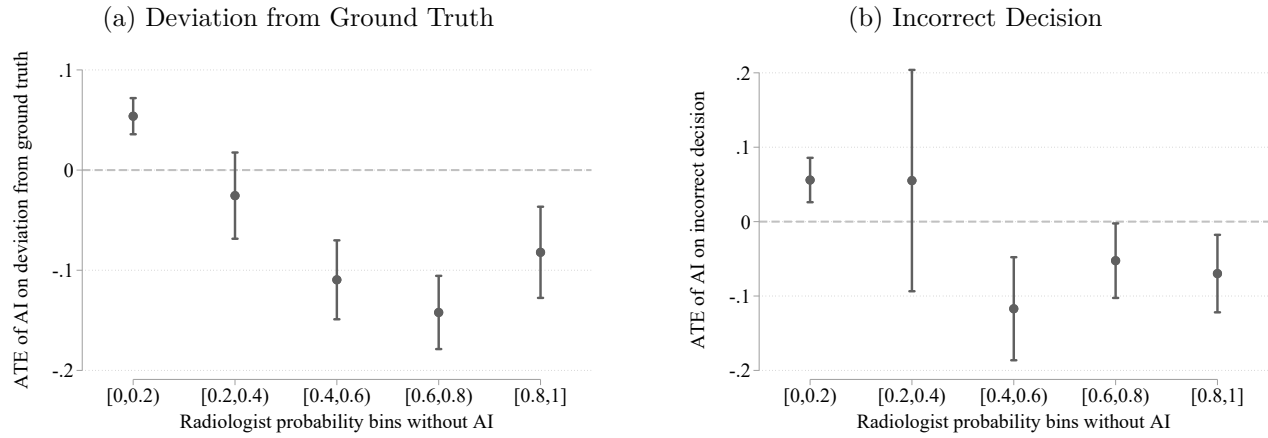
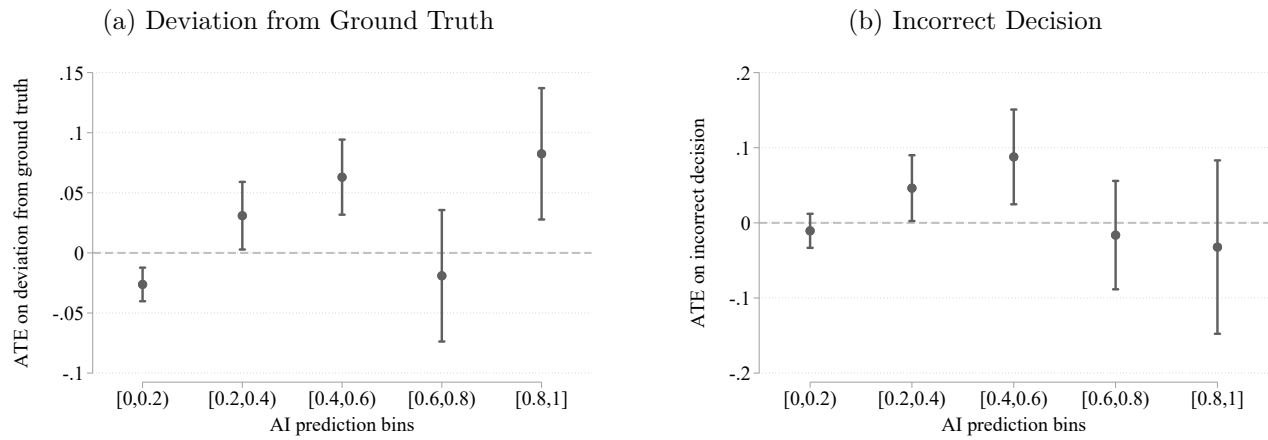


Figure C.18: Conditional treatment effect given radiologist prediction



Note: Main specifications similar to figure 2 with the exception that observations are from design 3 only.

Figure C.19: Conditional treatment effect given AI prediction



Note: Main specifications similar to figure 3 with the exception that observations are from design 3 only.

Table C.10: Average Treatment Effects

|                | Deviation<br>from AI | Deviation from<br>Ground Truth |
|----------------|----------------------|--------------------------------|
| Treatment      | (1)                  | (2)                            |
| AI $\times$ CH | 0.009<br>(0.008)     | 0.010<br>(0.012)               |
| AI             | -0.050<br>(0.010)    | -0.003<br>(0.008)              |
| CH             | -0.003<br>(0.009)    | -0.010<br>(0.013)              |
| CONTROL MEAN   | 0.241<br>(0.011)     | 0.216<br>(0.015)               |
| PATHOLOGY FE   | Yes                  | Yes                            |
| OBSERVATIONS   | 7000                 | 7000                           |

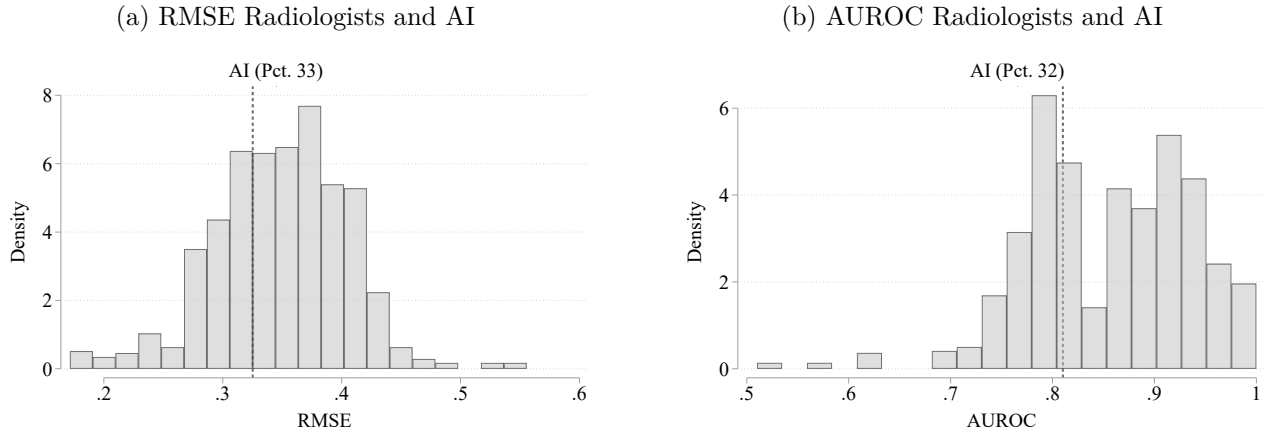
Note: Main specifications similar to table 2 with the exception that observations are from design 3 only.

### C.4.2 By Definition of Ground Truths

#### Experiment leave-one-out Ground Truth

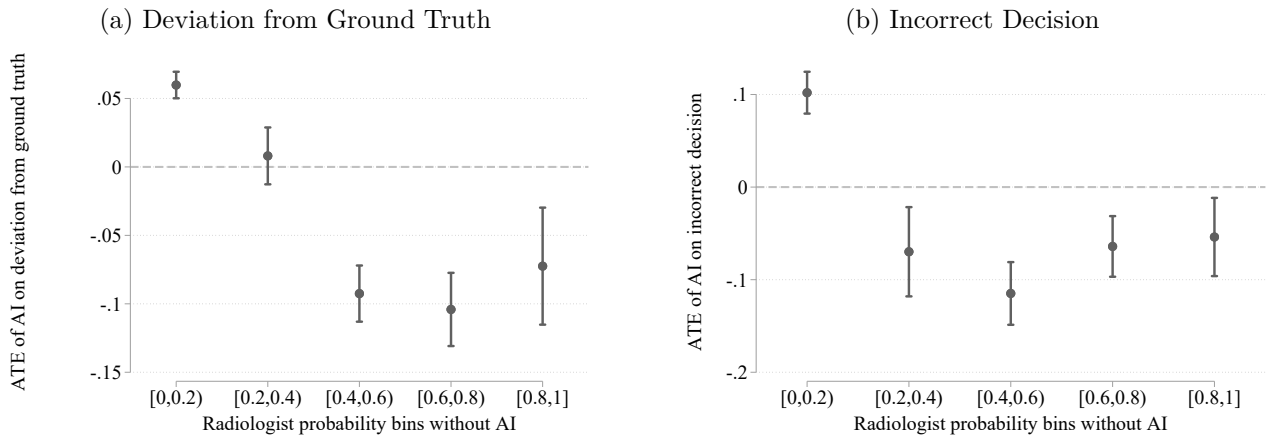
This section computes the main results using a ground truth constructed using a leave-one-out average of assessments by radiologists participating in the experiment in the treatment arm with clinical history but no AI assistance. Specifically, for each radiologist  $r$  and patient case  $i$  we construct  $\omega_{ir} = 1 \left[ \sum_{r' \neq r} \frac{\pi(\omega_i=1|s_{ir}^E)}{N_i-1} > 0.5 \right]$ .

Figure C.20: Comparing AI performance to Radiologists



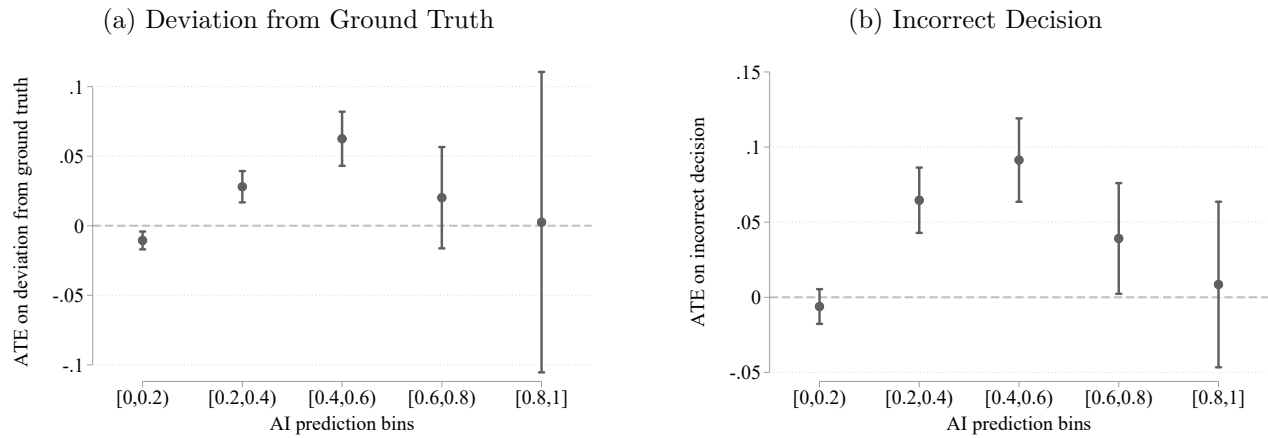
Note: Main specifications similar to figure 1 with the exception that ground truth is constructed using experiment leave-one-out average.

Figure C.21: Conditional treatment effect given radiologist prediction



Note: Main specifications similar to figure 2 with the exception that ground truth is constructed using experiment leave-one-out average.

Figure C.22: Conditional treatment effect given AI prediction



Note: Main specifications similar to figure 3 with the exception that ground truth is constructed using experiment leave-one-out average.

Table C.11: Average Treatment Effects

| Deviation from Ground Truth |                   |
|-----------------------------|-------------------|
| Treatment                   | (1)               |
| AI $\times$ CH              | 0.009<br>(0.005)  |
| AI                          | 0.007<br>(0.004)  |
| CH                          | -0.016<br>(0.004) |
| CONTROL MEAN                | 0.222<br>(0.010)  |
| PATHOLOGY FE                | Yes               |
| OBSERVATIONS                | 36280             |

Note: Main specifications similar to table 2 with the exception that ground truth is constructed using experiment leave-one-out average.